

Math 472: Homework 10

Due Wednesday, April 29

Problem 1 (Exercise 10.91). Let $Y_1, \dots, Y_{20}$ be a random sample of size $n = 20$ from a normal distribution with unknown mean and known variance $\sigma^2 = 5$. We wish to test $H_0 : \mu = 7$ versus $H_a : \mu > 7$.

- (a) Find the uniformly most powerful test with significance level .05.
- (b) For the test in part (a), find the power at each of the following alternative values for μ : $\mu_a = 7.5, 8.0, 8.5$, and 9.0.
- (c) Sketch a graph of the power function.

Problem 2. Let $X_1, \dots, X_n$ be a random sample from the pdf

$$f(x|\theta) = \theta x^{-2}, \quad 0 < \theta \leq x < \infty.$$

- (a) What is a sufficient statistic for θ ?
- (b) Find the MLE of θ .

Problem 3. In a given city it is assumed that the number of auto accidents in a given year follows a Poisson distribution. In past years, the average number of accidents per year was 15 and this year it was 10. Is it justified to claim that the accident rate has dropped?

Problem 4 (Exercise 10.21). Shear strength measurements derived from unconfined compression tests for two types of soils gave the results shown in the following table (measurements in tons per square foot, with s the sample standard deviation).

Soil Type	n	$\bar{y}$	s
I	30	1.65	0.26
II	35	1.43	0.22

Do the soils appear to differ with respect to average shear strength, at the 1% significance level? [Hint: see example 10.7 in the textbook.]

Problem 5. Refer to Problem 4. Construct a 99% confidence interval for the difference in mean shear strengths for the two soil types. [Hint: see example 8.8 in the textbook.]

- (a) Is the value $\mu_1 - \mu_2 = 0$ inside the interval?
- (b) Based on the interval, should the null hypothesis of Problem 4 be rejected? Why?
- (c) How does the conclusion compare with your conclusion Problem 4?